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VayuNetra:AI-Powered Carbon Footprint &Emission Intelligence Platform

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Abstract

VayuNetra is an AI-driven platform focused on environmental sustainability, helping people and communities track, study, and cut down on carbon emissions, especially those linked to transportation. It works in the area of AI for environmental monitoring by using rule-based models for calculating CO2 emissions, natural language processing, and location data to figure out personal carbon footprints and offer practical tips for eco-friendly living. The platform supports sustainable behavior through a digital system that rewards users with carbon credits, manages user roles, and provides interactive dashboards to show progress at both individual and community levels. By combining real-time environmental data with AI predictions about pollution spikes, VayuNetra helps users make smarter decisions that reduce their environmental impact. Its goal is to increase awareness, foster teamwork, and create real change toward a cleaner, more sustainable world.

Sustainable Development Goals(SDG) mapped

- SDG 11: Sustainable Cities and Communities Because vayunetra encourages eco-friendly transportation, promotes sustainable mobility, and supports cleaner, healthier communities.
- SDG 13: Climate Action Because the core aim of VayuNetra is to reduce carbon emissions, raise awareness, and fight against climate change through actionable insights and community engagement.

Introduction

- VayuNetra is a web-based platform that helps individuals and communities track and reduce carbon emissions from transportation.
- It calculates a person's carbon footprint using factors like travel mode, distance, fuel type, and vehicle-type, and then provides eco-friendly suggestions, carbon credits, and dashboards to encourage sustainable habits..

Motivation

- We were encouraged to do this project because transportation is one of the biggest contributors to pollution and climate change, and most people are unaware of their personal impact.
- Our goal is to create a system that makes people aware of their emissions and motivates them to adopt greener choices in daily life.

Objectives

- Use a rule-based model to calculate each user's carbon footprint, offer real-time sustainability tips using AI and natural language processing, and reward eco-friendly travel through a digital carbon credit system.
- Manage users with different access levels for employees, employers, banks, and admins; store and display carbon credits in secure digital wallets; and encourage environmental awareness with emission trends, rewards based on behavior, and community progress dashboards.
- Track group impact by collecting user data into community goals, showing progress, and using AI trained on travel and activity patterns to predict future emissions.
- To determine where the user is, you can use GPS or IP location tracking. Check if they are close to major industrial zones. Use pollution data from reliable sources or fixed datasets to assess carbon monoxide levels, particularly during nighttime industrial operations.
- Offer personalized insights by comparing the user's carbon savings with the emissions from nearby industries. Use emotional messages and visuals to highlight their impact and using AI to predict potential pollution increases.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	AI Driven Strategies For Carbon Footprint Reduction—A Scholarly Exploration[1]	Soumit Roy, Mainak Mitra	2020	AI-driven framework using ML & RNNs with IoT/sensor data for emission reduction	High data needs, high computation cost, deployment challenges
2	Evaluating the carbon footprint of NLP methods: a survey and analysis of existing tools.[2]	Nesrine Bannour, Sahar Ghannay, Aurélie Névéol, Anne-Laure Ligozat	2021	Survey & evaluation of six tools (Carbon Tracker, Experiment Impact Tracker, Green Algorithms, ML CO2 Impact, Energy Usage, Cumulator) on NER experiments	Tools underestimate emissions (ignore hardware production & lifecycle), results vary greatly, some tools lack precision for small experiments
3	AI Carbon Emissions Tracking of Machine Learning Models as the First Step Towards Sustainable AI[3]	S. A. Budennyy	2022	Introduces open source Python tool eco2AI, tested on large models DALL E, GPT, etc.).	Only tracks training/inference, not embodied or full lifecycle emissions.
4	How to estimate carbon footprint when training deep learning models? A guide and review[4]	Lucía Bouza	2023	Comparative review and experimental evaluation of AI carbon tracking tools.	Tool variance, focus mostly on training; not always process specific

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
5	A Review of IoT Based Visualisation Platforms for Tracking Household Carbon Footprints.[5]	Lanre Olatomiwa	2023	Literature review of IoT visualization platforms, scenario analysis.	Focused mainly on household platform, lacks empirical comparison.
6	Towards Sustainable AI Monitoring and Analysis of Carbon Emissions in Machine Learning Algorithms[6]	Aurora Martiny	2023	Empirical experiments, parameter studies on LSTMs, evaluation with CodeCarbon.	Focus on time-series, smaller scale, single location, tool limitations
7	Artificial intelligence applications in reduction of carbon emissions: Step towards sustainable environment[7]	Bhatti	2024	Editorial overview of recent AI approaches in emission reduction, literature context.	Lacks primary data or direct empirical evaluation, mostly summary.
8	AIDriven Carbon Footprint Tracking and Emission Reduction in Logistics Networks.[8]	VenkateshPrabu Parthasarathy	2024	Simulation, AI/ML/IoT-based tracking with case study logistics network, comparison to baseline.	Simulation—not real deployments; high initial setup costs; data quality sensitivity.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
9	A novel approach to sustainable behavior enhancement through AI driven carbon footprint assessment and real-time analytics.[9]	Ahmad Jasim Jasmy et al.	2024	Empirical study of mobile app for real time tracking, behavioral intervention, TRA model analysis	Small user sample, attitudes/intentions effect not significant.
10	Carbon Footprint of Artificial Intelligence AI Models: Estimation and Reduction Approaches.[10]	Mina Vildan	2025	Literature review, case studies, expert interviews.	Lack of standardized reporting, incomplete data, measurement inconsistency
11	Carbon Footprint Management in Global Supply Chains: A Data-Driven Approach Utilizing Artificial Intelligence.[11]	Rong Huang, Shuai Mao	2025	Data-driven framework using AI (random optimization, CNN deep learning) for emissions prediction & supply chain optimization; simulation-based analysis for emission reduction	Strong reliance on data availability & quality; high computational cost; limited interpretability for stakeholders
12	India Specific Road Transport Emission Factors	WRI India, TERI, CII	2015	Developed Indian emission factors for passenger, freight, and bus transport using the formula: **Emission Factor (kg CO_2/km) = (Fuel Emission Factor (kg CO_2/litre) ÷ Fuel Efficiency (km/litre))**, adjusted with uplift factor (≈8–10%) for real-world driving conditions.	Lacks granular, region-specific vehicle data and real-time emission tracking; assumes constant efficiency and uplift.

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	Technical Guidance for Calculating Scope 3 Emissions (Category 6: Business Travel)	World Resources Institute (WRI), GHG Protocol	2013	Provides three methods for calculating transport emissions — **Fuel-based, Distance-based, and Spend-based** — with the distance method: $CO_2e = \Sigma(di \times EFi)$ over all travel modes.	Designed for corporate reporting, not individual real-time monitoring; limited personal-level insights.
14	COPERT II: Road Transport Emission Calculation Methodology	European Environment Agency (EEA), Ntziachristos & Samaras	1997	Establishes the COPERT model for road traffic emissions using vehicle categories, speed-dependent emission factors, and correction parameters (road gradient, weight, climate).	High data and computation requirements; developed for European context, not directly transferable to Indian driving conditions.

Calculations

Formula: $CC = \sum[(EF_{baseline} - EF_{actual}) \times Distance \times Time_Weight \times Context_Factor]$

- CC = Carbon Credits earned by user
- EFbaseline = The emission factor representing a standard or default emission rate per unit distance (e.g., if a user always drove a petrol car)
- EFactual = The actual emission factor of the user's chosen commute mode (e.g., electric bike, carpool, hybrid car)
- Distance = The distance traveled during the commute
- Time_Weight = A weighting factor to account for time-based factors, such as peak vs off-peak hours, or the duration/frequency of commute (longer, more frequent commutes might have different weighting)
- Context_Factor = An adjustment reflecting environmental context (e.g., pollution levels nearby, route traffic conditions, weather, or industrial proximity)

Calculations

- Time: Peak evening, Tuesday Weather: Favorable conditions, city center routeStep-by-Step Calculation:
- Step 1: Emission Factors
- EF_baseline: 0.12 kg CO_2/km (cycling baseline)
- EF_actual: 0.00 kg CO_2/km (zero emissions)
- Savings: 0.12 kg CO_2/km
- Step 2: Distance
- Distance: 17.55 km
- Duration: 54.7 minutes
- Speed: 19.2 km/h
- Step 3: Time Weight
- Time_Weight = Peak_Factor × Traffic_Multiplier × Recency_Weight
- = 1.2 × 1.0 × 1.0 = 1.2
- Peak evening (1.2) + Light traffic (1.0) + Recent (1.0)
- Step 4: Context Factor
- Context_Factor = Weather × Route × Load × AQI × Seasonal
- = 0.95 × 1.2 × 1.0 × 1.1 × 0.95 = 1.254
- Favorable weather (0.95) + City center (1.2) + Poor AQI (1.1) + Winter (0.95)

Step 5: Final Calculation

$CC = (0.12 - 0.00) \times 17.55 \times 1.2 \times 1.254$CC = $0.12 \times 17.55 \times 1.2 \times 1.254$CC = 3.1691 kg CO_2

🎯 Result: This cycling trip earned 3.17 kg CO_2 credits - equivalent to avoiding 24.4 km of petrol car travel!

Research Gap(Limitations of existing systems)

- Only measure active energy use, ignoring emissions from hardware production, idle use, and disposal.
- Regional variations in emission factors are inconsistent and inaccurate results.
- Lack sensitivity for small or short experiments, relying on rough estimates.
- Real-time trackers need complex permissions; online tools require difficult input data.
- No standardization, making cross-tool comparisons unreliable.
- Privacy & security concerns in handling sensitive usage data.
- High setup and maintenance costs for IoT/tracking systems.
- Compatibility issues across devices and platforms.
- Low user awareness and high technical barriers hinder adoption.

Problem Definition

The rapid growth of transportation has led to rising carbon emissions, causing climate change, pollution, and health risks.Existing systems like PUC only give periodic checks and fail to provide continuous, actionable insights into vehicle emissions.There is also no transparent system to motivate people to adopt greener travel.VayuNetra fills this gap with an AI-powered platform that monitors carbon footprints and rewards eco-friendly practices.

Scope

1. Development of an AI-powered platform for calculating and analyzing vehicle carbon footprints using fuel consumption and travel data.
2. Provision of actionable insights and reports for individual users, employers, banks, and admins.
3. Design and implementation of a Carbon Credit System that rewards users for adopting eco-friendly driving practices and low-emission vehicles.
4. Creation of an Emission Intelligence Dashboard for clear and meaningful visualization of carbon footprint data.
5. Promotion of sustainable mobility choices such as electric vehicles, fuel-efficient driving, and shared transport through incentives.

Technological Stack

❖ Carbon Footprint Calculation (Rule-Based CO_2 Emission Model)

- Technology Stack: Python (Django), SQLite, Pandas/Numpy
- Method: Use rule-based formulas with multiple emission
- factors:Transport mode × distance × fuel typeVehicle occupancy/load factor, Travel frequency & time of day (day vs. night), Proximity to industrial zones
- Customization: Provides more accurate, personalized footprint estimates compared to simple mode–distance models.

❖ AI-Powered Sustainability Insights (NLP + Predictive Analytics)

- Technology Stack: Python (scikit-learn/TensorFlow optional), Django REST Framework, NLP libraries (spaCy/HuggingFace)
- Method: NLP to answer user queries.
- Predictive analysis on travel patterns to forecast future emissions.
- Customization: Delivers real-time eco-friendly tips (walk, cycle, carpool, public transport) and projected emission trends.

Technological Stack

❖ Digital Carbon Credit & Wallet System

- Technology Stack: Django Models, REST APIs, SQLite
- Method: Eco-friendly actions, converted into carbon credits, stored in secure digital wallets.
- Customization: Rewards are strictly for employees and employers. Redeemable as vouchers, recognition badges, or company-driven incentives.

❖ Community Progress & Dashboards

- Technology Stack: HTML, Tailwind CSS, Chart.js / D3.js, Django Templates
- Method: Aggregate user-level data shows company/community sustainability progress.
- Customization: Leaderboards, progress bars, and historical trends to motivate collective action.

Technological Stack

❖ Location & Pollution Awareness Module

- Technology Stack: Geolocation APIs (GPS/IP), pollution datasets/APIs, Python (pandas)
- Method: Detect user location, assess proximity to industrial zones, and map against real-time/recorded pollution levels.
- Customization: Emotional visualizations ("Your weekly savings offset the pollution from 2 factory hours") + alerts for pollution surges (e.g., nighttime industrial activity).

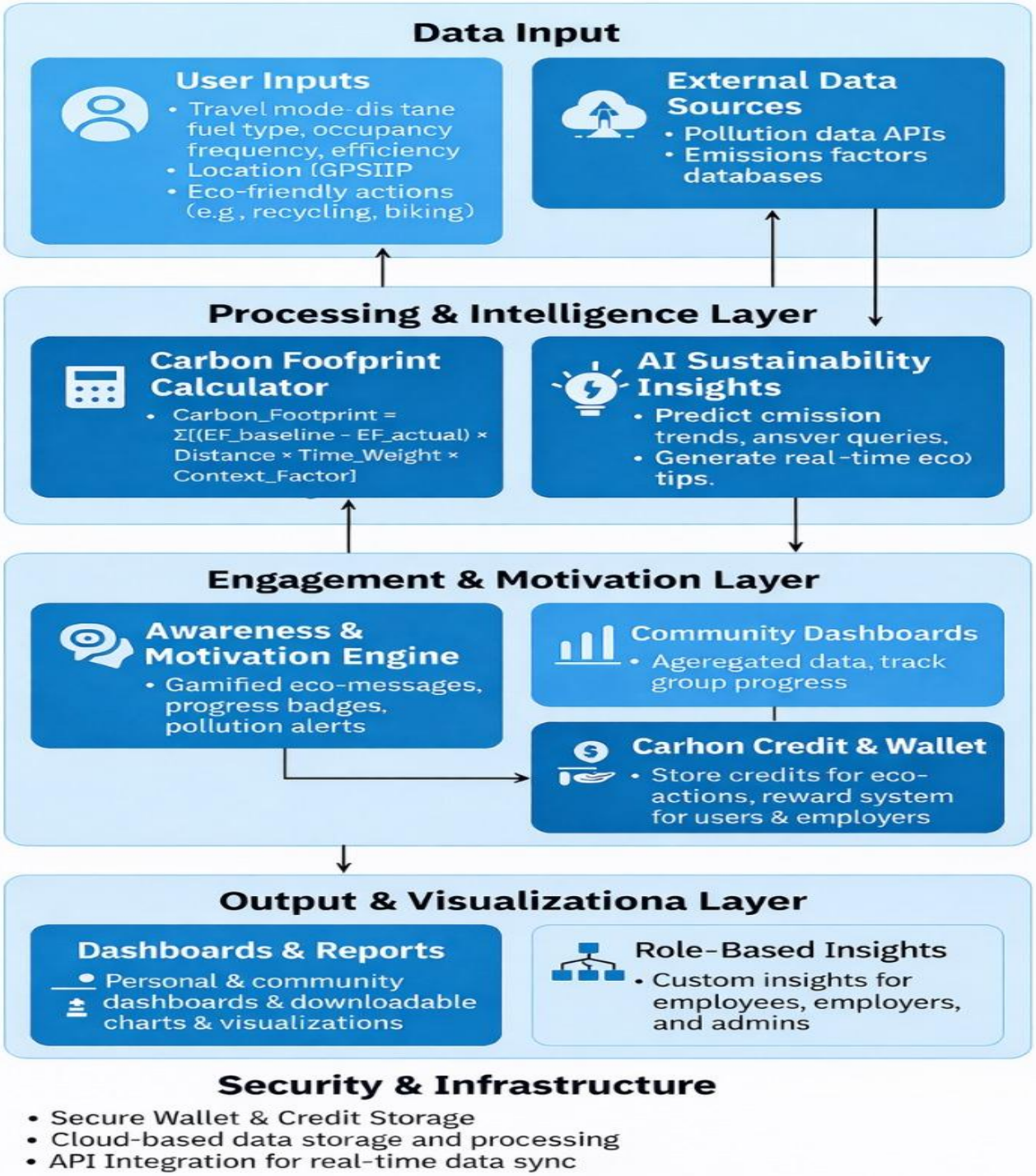
❖ Awareness & Motivation Engine

- Technology Stack: Django + Tailwind CSS for UI
- Method: Deliver gamified eco-messages and comparisons (e.g., "You saved emissions equal to planting 5 trees ").
- Customization: Positive reinforcement through personalized messages and progress milestones.

❖ System Deployment & Integration:

1. Frontend: HTML, Tailwind CSS.
2. Backend: Django
3. Database: SQLite
4. Deployment: Render cloud services
5. Customization: Unified web application integrating carbon tracking, AI tips, wallet system, dashboards, and pollution insights.

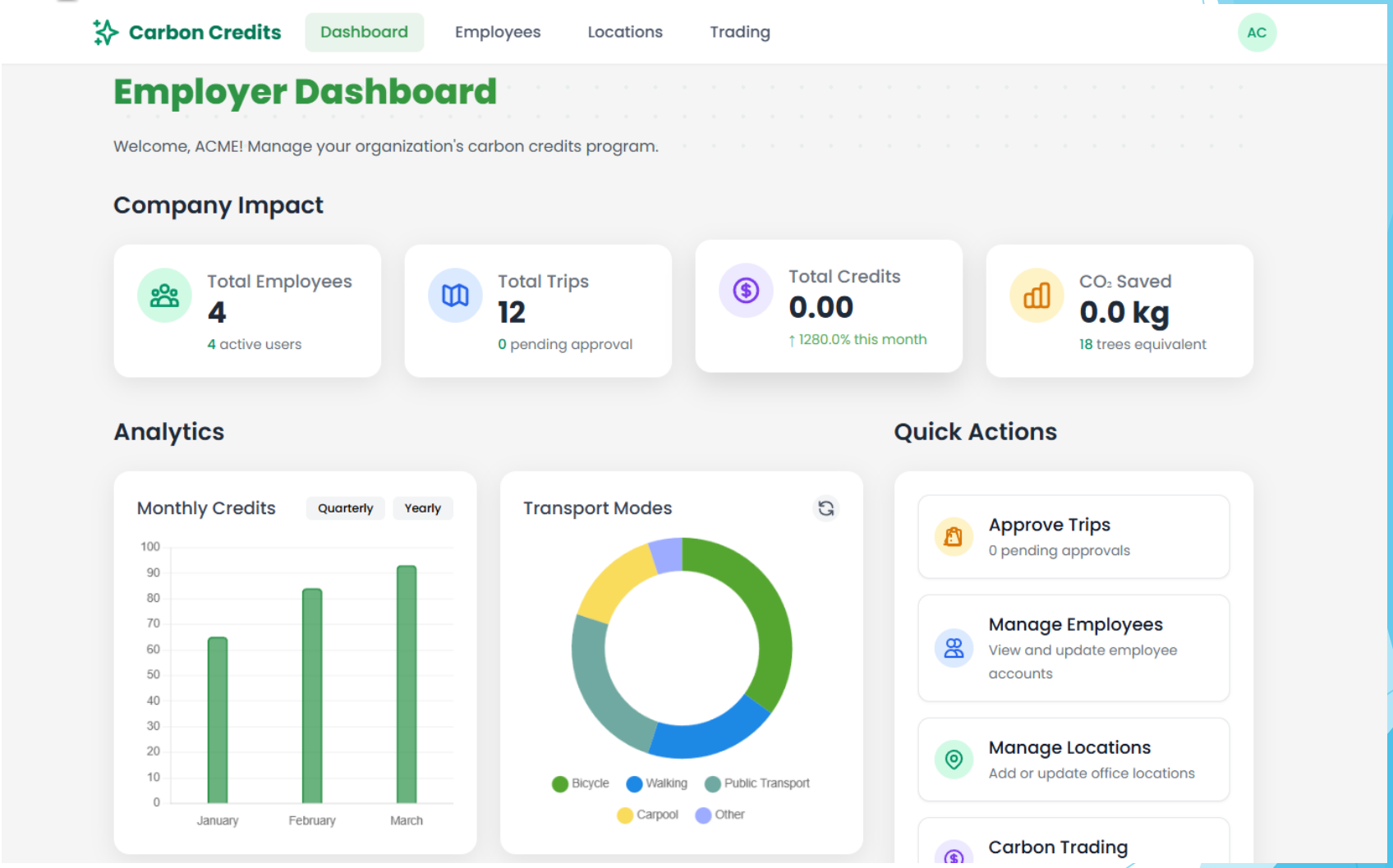
Proposed system architecture/Working/Module Diagram



Prototype Design Demonstration

Not Applicable for Review 1

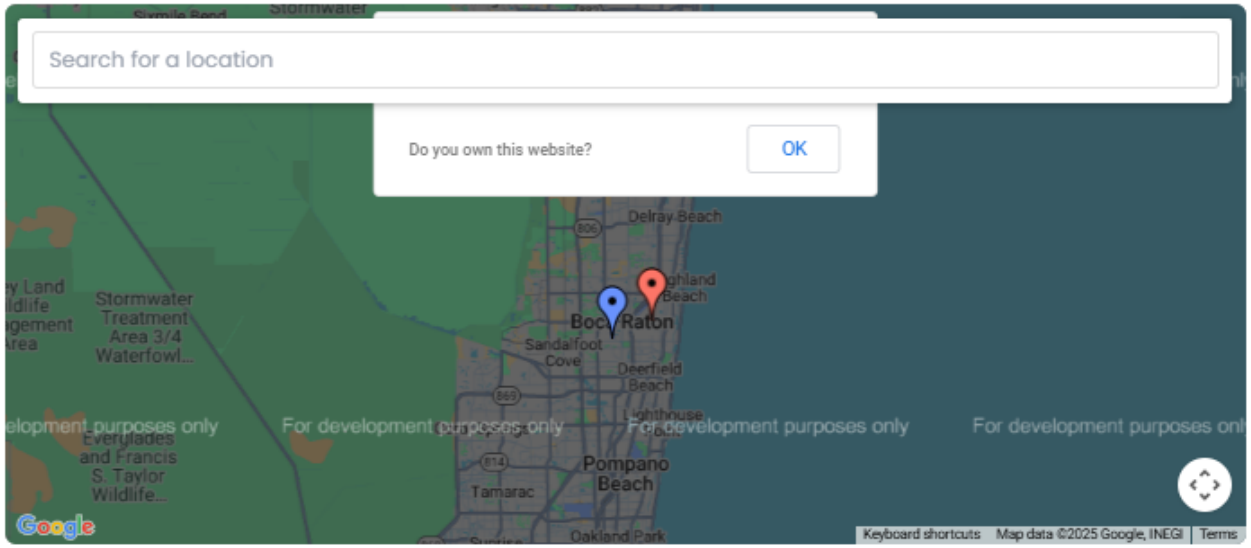
Implementation Status



Carbon Credits | Dashboard | Log Trip | My Trips | JD

Trip Route

The map shows the route between your selected locations. Distance and travel time will be calculated based on your transport mode.



Trip Date

25-08-2025

Transport Mode

Select how you commuted. Different modes earn different amounts of carbon credits.

Public Transport
+3 credits/km

Bicycle
+5 credits/km

Walking
+6 credits/km

Carpool
+2 credits/km

Car (Single)
+0.5 credits/km

Work from Home
+10 credits

Trip Proof

Upload an image of your ticket, receipt, or any proof of your commute.

Advanced Upload — With Preview

Quick Upload — Choose File

Review Suggestions (Given in Last meeting)

- Keep Scope till Private cars/bikes , cabs and rikshaws and not huge vehicle like bus to ease the calculations.
- Need to refer more papers to determine the calculations.
- Look for datasets that can help to determine the formulations
- Modify the architecture Diagram.

Implementation Status

Last semester, we completed two objectives. In this semester, we have completed about 85% of the project.

Review Suggestions (Given in Last meeting)

Comments Given During External Viva (Oct/Nov 2025)	Solutions Implemented
1] Focus on formula validation	Referred to WRI India and IPCC standards for formula validation.
2] Expand the dataset	Expanded the dataset using validated formulas.
3] Identify pollution hotspot areas	Used Google Maps API to display real-time polluted areas based on AQI data.

Research paper Status

The research paper has been completed, with 3% similarity and 0% AI plagiarism.

Patent/Copyright filing Status

References

- [1] Soumit Roy and Mainak Mitra, AI Driven Strategies for Carbon Footprint Reduction – A Scholarly Exploration, International Journal of Current Science (IJCSPUB), Volume 10, Issue 1, pp. 15–20, 2020. https://www.researchgate.net/publication/375957421
- [2] Nesrine Bannour, Sahar Ghannay, Aurélie Névéol and Anne-Laure Ligozat, Evaluating the Carbon Footprint of NLP Methods: A Survey and Analysis of Existing Tools, Proceedings of the 2nd Workshop on Simple and Efficient Natural Language Processing (SustaiNLP), Association for Computational Linguistics, pp. 11–21, 2021. https://doi.org/10.18653/v1/2021.sustainlp-1.2
- [3] Semen Budennyy, Vladimir Lazarev, Nikita Zakharenko, Alexey Korovin, Olga Plosskaya, Denis Dimitrov, Vladimir Arkhipkin, Ivan Oseledets, Ivan Barsola, Ilya Egorov, Aleksandra Kosterina, Leonid Zhukov, Eco2AI: Carbon Emissions Tracking of Machine Learning Models as the First Step Towards Sustainable AI, 2022, [https://arxiv.org/abs/2208.00406]
- [4] Lucía Bouza, Aurélie Bugeau and Loïc Lannelongue, How to estimate carbon footprint when training deep learning models? A guide and review, Environmental Research Communications, Volume 5, Article 115014, IOP Publishing Ltd., 2023. https://doi.org/10.1088/2515-7620/acf81b

- [5] Lanre Olatunde Olatomiwa, James Garba Ambafi, Umar Suleiman Dauda, et al., A Review of Internet of Things-Based Visualization Platforms for Tracking Household Carbon Footprints, Sustainability, Volume 15, Article 15016, 2023, DOI: 10.3390/su152015016.
- [6] Aurora Martiny, Towards Sustainable AI: Monitoring and Analysis of Carbon Emissions in Machine Learning Algorithms, Master's Thesis, Politecnico di Torino, 2023.
- [7] Uzair Aslam Bhatti, Muhammad Zaffar Hashmi, Yuzhen Sun, Mehedi Masud, Mir Muhammad Nizamani, Artificial Intelligence Applications in Reduction of Carbon Emissions: Step Towards Sustainable Environment, Frontiers in Environmental Science, Volume 11, Article 1183620, 2023, DOI: 10.3389/fenvs.2023.1183620, https://www.frontiersin.org/articles/10.3389/fenvs.2023.1183620/full
- [8] VenkateshPrabu Parthasarathy "AI-Driven Carbon Footprint Tracking and Emission Reduction in Logistics Networks" ,2024, https://www.intangles.ai/ai-powered-carbon-tracking-the-key-to-sustainable-logistics/
- [9] Ahmad Jasim Jasmy, Heba Ismail, Noof Aljneibi, A Novel Approach to Sustainable Behavior Enhancement Using AI-Driven Carbon Footprint Assessment and Real-Time Analytics, Environment, Development and Sustainability, 2024, DOI: 10.1007/s43621-024-00762-w.

- [10] Mina Vildan Şark, Carbon Footprint of Artificial Intelligence (AI) Models: Estimation and Reduction Approaches, Master's Final Dissertation, Master in Innovation and Research for Sustainability, February, 2025 https://www.advancedsciencenews.com/calculating-the-true-environmental-costs-of-ai/
- [11] Rong Huang and Shuai Mao, Carbon Footprint Management in Global Supply Chains: A Data-Driven Approach Utilizing Artificial Intelligence, *2024*. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10547105
- [12] WRI India; TERI; CII, India-Specific Road Transport Emission Factors, 2015. (Method report with CO_2 per km factors for Indian road transport.)
- [13] World Resources Institute (WRI) & GHG Protocol, Technical Guidance for Calculating Scope 3 Emissions – Category 6: Business Travel, 2013. (Fuel-based, distance-based, spend-based methods.)
- [14] Leonidas Ntziachristos and Zissis Samaras, COPERT II—Computer Programme to Calculate Emissions from Road Transport: Methodology and Emission Factors, European Environment Agency, 1997.

Thank You...!!